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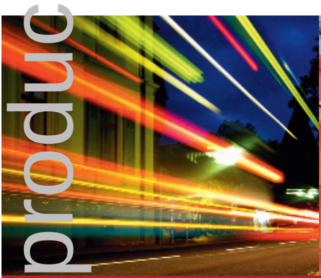
BS EN 50388:2012

Incorporating corrigenda August 2012, April 2013 and August 2013





BSI Standards Publication

Railway Applications — Power supply and rolling stock — Technical criteria for the coordination between power supply (substation) and rolling stock to achieve interoperability

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National foreword

This British Standard is the UK implementation of EN 50388:2012, incorporating corrigenda August 2012, April 2013 and August 2013. It supersedes BS EN 50388:2005 which is withdrawn.

The UK participation in its preparation was entrusted by Technical Committee GEL/9, Railway Electrotechnical Applications, to Subcommittee GEL/9/3, Railway Electrotechnical Applications – Fixed Equipment.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

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Compliance with a British Standard cannot confer immunity from legal obligations.

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Date	Text affected
31 October 2012	Implementation of CENELEC corrigendum August 2012: Figure 1 updated
31 May 2013	Implementation of CENELEC corrigendum April 2013: Figure 1 replaced and Clause 12.1.1 amended
30 September 2013	Implementation of CENELEC corrigendum August 2013: Table F.1, Column: SK and Row: d.c. 3000 V amended

EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 50388

March 2012

ICS 29.280; 45.060.01

Incorporating corrigenda August 2012, April 2013 and August 2013

English version

Railway Applications - Power supply and rolling stock - Technical criteria for the coordination between power supply (substation) and rolling stock to achieve interoperability

Applications ferroviaires -
Alimentation électrique
et matériel roulant -
Critères techniques pour la coordination entre le système d'alimentation (sous-station) et le matériel roulant pour réaliser l'interopérabilité

Bahnanwendungen -
Bahnenergieversorgung und Fahrzeuge -
Technische Kriterien für die Koordination zwischen Anlagen der Bahnenergieversorgung und Fahrzeugen zum Erreichen der Interoperabilität

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European Committee for Electrotechnical Standardization
Comité Européen de Normalisation Electrotechnique
Europäisches Komitee für Elektrotechnische Normung

Management Centre: Avenue Marnix 17, B - 1000 Brussels

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Foreword

This document (EN 50388:2012) has been prepared by CLC/SC 9XC, "Electric supply and earthing systems for public transport equipment and ancillary apparatus (Fixed installations)", of Technical Committee CLC/TC 9X, "Electrical and electronic applications for railways". It also concerns the expertise of CLC/SC 9XB, "Electromechanical material on board of rolling stock".

The following dates are fixed:

• latest date by which this document has to be implemented at national level by publication of an identical national standard or by endorsement	(dop)	2013-02-13
• latest date by which the national standards conflicting with this document have to be withdrawn	(dow)	2015-02-13

This document supersedes EN 50388:2005 + corrigendum May 2010.

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For the relationship with EU Directive 2008/57/EC, see informative Annex ZZ, which is an integral part of this document.

For TSI lines, modification and amendments shall be made within a procedure which is related to the legal status of the HS and CR TSIs.

1 Scope

This European Standard establishes requirements for the compatibility of rolling stock with infrastructure particularly in relation to:

- co-ordination of protection principles between power supply and traction units, especially fault discrimination for short-circuits;
- co-ordination of installed power on the line and the power demand of trains;
- co-ordination of traction unit regenerative braking and power supply receptivity;
- co-ordination of harmonic behaviour.

This European Standard deals with the definition and quality requirements of the power supply at the interface between traction units and fixed installations.

This European Standard specifies the interface between rolling stock and electrical fixed installations for traction, in respect of the power supply system. The interaction between pantograph and overhead contact line is dealt with in EN 50367. The interaction with the “control-command” subsystem (especially signalling) is not dealt with in this standard.

Requirements are given for TSI lines (both high speed and conventional) and classical lines.

For classical lines, values, where given, are for the existing European networks. Furthermore the maximum values that are specified are applicable to the foreseen developments of the infrastructure of the Trans European rail networks.

The following electric traction systems are within scope:

- railways;
- guided mass transport systems that are integrated with railways;
- material transport systems that are integrated with railways.

This European Standard does not apply retrospectively to rolling stock already in service.

Information is given on electrification parameters such as to enable train operating companies to confirm, after consultation with the rolling stock manufacturers, that there will be no consequential disturbance on the electrification system.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 50122-2:2010, *Railway applications — Fixed installations — Electrical safety, earthing and the return circuit — Part 2: Provisions against the effects of stray currents caused by d.c. traction systems*

EN 50122-3:2010, *Railway applications — Fixed installations — Electrical safety, earthing and the return circuit — Part 3: Mutual Interaction of a.c. and d.c. traction systems*

EN 50123-1:2003, *Railway applications — Fixed installations — D.C. switchgear — Part 1: General*

EN 50163:2004 + A1:2007, *Railway applications — Supply voltages of traction systems*

EN 50367, *Railway applications — Current collection systems — Technical criteria for the interaction between pantograph and overhead line (to achieve free access)*

IEC 60050-811, *International Electrotechnical Vocabulary — Chapter 811: Electric traction*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
abnormal operating conditions
either higher traffic loads or outage of power supply equipment outside the design standard

Note 1 to entry: Under these conditions, traffic may not operate to the design timetable.

3.2
classical line
existing line that is not subject to a renewal or upgrading project to bring it into compliance with a TSI

3.3
contact line
conductor system for supplying electric energy to vehicles through current-collecting equipment

[SOURCE: IEC 60050-811-33-01]

3.4
dimensioning train
train with the lowest mean useful voltage

3.5
infrastructure manager
body or undertaking that is responsible in particular for establishing and maintaining railway infrastructure

Note 1 to entry: This may also include the management of infrastructure control and safety systems. The functions of the infrastructure manager on a network or part of a network may be allocated to different bodies or undertakings.

Note 2 to entry: In TSI Energy, this body is referred to as the contracting or adjudicating entity.

3.6
maximum line speed
speed for which the line was approved for operation

3.7
mean useful voltage at the pantograph ($U_{\text{mean useful}}$)

3.7.1
$U_{\text{mean useful}}$ (zone)
voltage giving an indication of the quality of the power supply in a geographic zone during the peak traffic period in the timetable

3.7.2
$U_{\text{mean useful}}$ (train)
voltage identifying the dimensioning train and which enables the effect on its performance to be quantified

3.8
minimum possible headway
minimum interval at which trains can run as allowed by the signalling system

3.9
new element
new, rebuilt or modified traction-unit or power supply component (hardware or software) having a possible influence on the harmonic behaviour of the power supply system

Note 1 to entry: This new element may be integrated in an existing power supply network with traction units e.g. for fixed installation:

- transformer;
- HV cable;
- filters;
- converter.

3.10
normal operating conditions
traffic operating to the design timetable and train formation used for power supply fixed installation design

Note 1 to entry: Power supply equipment is operated according to standard design rules. Rules can vary depending on the infrastructure manager's policy.

3.11
overhead contact line
contact line placed above (or beside) the upper limit of the vehicle gauge and supplying vehicles with electric energy through roof–mounted current collection equipment

[SOURCE: IEC 60050-811-33-02]

3.12
power factor

$$\cos\varphi = \frac{active\ power\ of\ the\ fundamental\ wave}{apparent\ power\ of\ the\ fundamental\ wave}$$

Note 1 to entry: In this standard, only the fundamental wave is considered.

Note 2 to entry: This is also the displacement factor cos φ.

3.13
register of infrastructure
for TSI, a single document which compiles, for each section of line, the characteristics of the lines concerned in respect of all subsystems including fixed equipment

Note 1 to entry: The list of items included in the register is described in annexes of the TSI Energy.

3.14
rolling stock
general term covering all vehicles with or without motors

[SOURCE: IEC 60050-811-02-01]

3.15
separation or neutral section
section of a contact line provided with a sectioning point at each end to prevent successive electrical sections, differing in voltage, phase or frequency being connected together by the passage of current collectors

3.16
(traction) substation
installation, the main function of which is to supply a contact line system, at which the voltage of a primary supply system, and in certain cases the frequency, is converted to the voltage and frequency of the contact line

3.17
total power factor λ

$$\lambda = \frac{active\ power}{apparent\ power}$$

Note 1 to entry: Deformation factor υ:

$$\upsilon = \frac{\lambda}{\cos \varphi}$$

3.18
traction unit
general term covering a locomotive, motor coach or train unit

[SOURCE: IEC 60050-811-02-04]

3.19
train
combination of rolling stock coupled together. It includes banking locomotives

3.20
train power at the pantograph
active power of the train taking into account power for traction, regeneration and auxiliaries

3.21
TSI line
high speed or conventional rail line being part of the Trans-European Rail Network (TEN) and complying with the requirements of the relevant Technical Specifications for Interoperability (TSI)

Note 1 to entry: It includes for the High Speed rail network:

– category I: specially built high-speed lines equipped for speeds generally equal to or greater than 250 km/h;
– category II: specially upgraded high-speed lines equipped for speeds of the order of 200 km/h;
– category III: specially upgraded high-speed lines which have special features as a result of topographical, relief or town planning constraints on which the speed must be adapted to each case.

It includes for the Conventional Rail network:

a) category IV: New Core Trans European Network (TEN) Line:
   1) passenger and mixed traffic: 200 km/h maximum;
   2) freight traffic: 140 km/h maximum;
b) category V: Upgraded Core TEN Line:
   1) passenger and mixed traffic: 160 km/h maximum;
   2) freight traffic: 100 km/h maximum;
c) category VI: New Other TEN Line:
   1) passenger and mixed traffic: 140 km/h maximum;
   2) freight traffic: 100 km/h maximum;
d) category VII: Upgraded Other TEN Line:
   1) passenger and mixed traffic: 120 km/h maximum;
   2) freight traffic: 100 km/h maximum

3.22
type of line
classification of lines as a function of the parameters described in 3.6, 3.8 and 3.20

3.23
vehicle
general term denoting any single item of rolling stock, e.g. a locomotive, a coach or a wagon

[SOURCE: IEC 60050-811-02-02]

4 Periods over which parameters can be averaged or integrated

Where train operators or infrastructure managers use various parameters for their dimensioning computations, protection measures and planning, these are effective only if they are averaged over precisely defined time spans. Guidance and recommendations on these time spans are given in Annex A (informative).

5 Separation sections

5.1 Phase separation sections

Trains shall be able to move from one section of an a.c. system to an adjacent section of the same system, through a phase separation section, without bridging the different phases.

Power consumption of the train (traction, auxiliaries and no-load current of the transformer) shall be brought to zero before entering the phase separation section.

For HS TSI lines, this shall be done automatically.

For Conventional Rail TSI lines and for classical lines, where required by the infrastructure manager this shall be done automatically. Otherwise, automatic operation is preferred, but manual on board operation may also be employed.

Where particular circumstances require the lowering of the pantographs this shall be recorded in the register of infrastructure.

For phase separation sections longer than 8 m, the infrastructure manager shall provide adequate means to allow a train that is gapped underneath the phase separation to be restarted.

EN 50367 describes the characteristics of some designs of phase separation sections.

NOTE For other designs of phase separation that allow trains to pass the section with power running (e.g. automatically switched sections or “change over sections”), the requirements of this clause may not apply if reliability and compatibility with all trains can be demonstrated.

5.2 System separation sections

5.2.1 General

Trains shall be able to move from one energy supply system to an adjacent one which uses a different energy supply without bridging the two contact line systems. The necessary actions (opening of the main circuit breaker, lowering of the pantographs, etc.) depend on the type of both supply systems as well as on the arrangement of pantographs on trains and the running speed.

There are two possibilities for trains to run through system separation sections:

1) with pantograph raised and touching the contact wire(s) as described in 5.2.2;

2) with pantograph lowered and not touching the contact wire(s) as described in 5.2.3.

The choice between 1) and 2) shall be made by the infrastructure manager.

The requirements for the design of the infrastructure and rolling stock are described in following paragraphs.

5.2.2 Pantograph raised

Where the system separation sections are traversed by a train with pantographs raised to the contact wire(s), the following conditions apply:

- Provisions shall be made in the infrastructure to avoid bridging the contact lines of both adjacent power supply systems if the opening of the on-board circuit breaker(s) fails.
- For categories I, II and III lines, on board, devices shall automatically open the circuit breaker before reaching the separation section and shall recognise automatically the voltage of the new power supply system at the pantograph in order to switch the corresponding circuits.
- For categories IV to VII lines and for classical lines, the requirements for categories I, II and III lines may be applied.

5.2.3 Pantograph lowered

Where the system separation sections are traversed with pantographs lowered the following conditions apply:

- The design of separation section between differing energy supply systems shall ensure that, in case of a pantograph unintentionally applied to the contact line, bridging the contact lines of two power supply systems is avoided and switching off both supply systems is triggered immediately, e.g. by detection of short circuits or unintended voltages.
- For categories I, II and III lines, at supply system separations which require a lowering of the pantograph, the pantograph shall be lowered without the driver’s intervention, triggered by control signals.
- For categories IV to VII lines and for classical lines, the requirements for categories I, II and III lines may be applied.
- EN 50367 describes the design of the system separation sections as well as other functional requirements of the overhead contact line and pantographs.

6 Power factor of a train

6.1 General

To optimise the power factor of a train, and hence the quality of the power supply performance, the requirements in 6.2 to 6.4 shall apply to the design of the train.

NOTE Capacitive or inductive power from a train can be utilised to change the contact line voltage.

6.2 Inductive power factor

This clause deals only with inductive power factor and power consumption over the range of voltage from U_{min1} to U_{max1} as defined in EN 50163.

Table 1 gives the requirements for the total inductive power factor λ of a train. For the calculation of λ, only the fundamental of the voltage at the pantograph is taken into account.

Table 1 — Total inductive power factor λ of a train

	Total inductive power factor λ of a train	
Instantaneous train power P at the pantograph (MW)	**Category I and II of HS TSI lines [a]**	**TSI line category III; IV; V; VI; VII and Classical lines**
$P > 2$	$\geq 0{,}95$	$\geq 0{,}95$
$0 \leq P \leq 2$	[b]	[b]

For yards or depots, the power factor of the fundamental wave shall be equal or higher to 0,8 (see Note 2 below) when the train is hotelling with traction power switched off and all auxiliaries running and the active power being drawn is greater than 200 kW.

NOTE 1 The calculation of overall average λ for a train journey, including the stops, is taken from the active energy W_P (MWh) and reactive energy W_Q (Mvarh) given by a computer simulation of a train journey or metered on an actual train:

$$\lambda = \sqrt{\frac{1}{1+\left(\frac{W_Q}{W_P}\right)^2}}$$

NOTE 2 Higher power factors than 0,8 will result in better economic performance due to a reduced requirement for fixed equipment provision.

[a] Applicable to trains in conformity with the HS TSI "rolling stock".

[b] In order to control the total power factor of the auxiliary load of a train during the coasting phases, the overall average λ (traction and auxiliaries) defined by simulation and/or measurement shall be higher than 0,85 over a complete timetable journey (typical journey between two stations including commercial stops).

During regeneration, the inductive power factor may be allowed to decrease freely in order to keep voltage within limits.

NOTE 1 Another representation of Table 1 in a graphic form is given in Annex E.

NOTE 2 On line categories III to VII, for rolling stock existing before publication of this standard, the infrastructure manager may impose conditions e.g. economic, operating, power limitation for acceptance of interoperable trains having power factors below the value specified in Table 1.

6.3 Capacitive power factor

During traction mode and standstill, requirements for capacitive power factor in order to keep voltage within limits are:

– within the range of voltage from U_{min1} to U_{max1} as defined in EN 50163, capacitive power factors are not limited;

– within the range of voltage from U_{max1} to U_{max2} as defined in EN 50163, a train shall not behave like a capacitor.

During regenerative mode, capacitive power, if any, shall be limited to 150 kvar within the range of voltage from U_{min1} to U_{max1} as defined in EN 50163.

NOTE 1 Capacitive power factors could lead to overvoltages and/or dynamic effects and should be treated according to Clause 10.

NOTE 2 A representation of capacitive power factor is given in Annex E.

6.4 Acceptance criteria

The power factor is acceptable if the values given in Table 1 and requirements given in 6.3 are achieved.

7 Train current limitation

7.1 Maximum train current

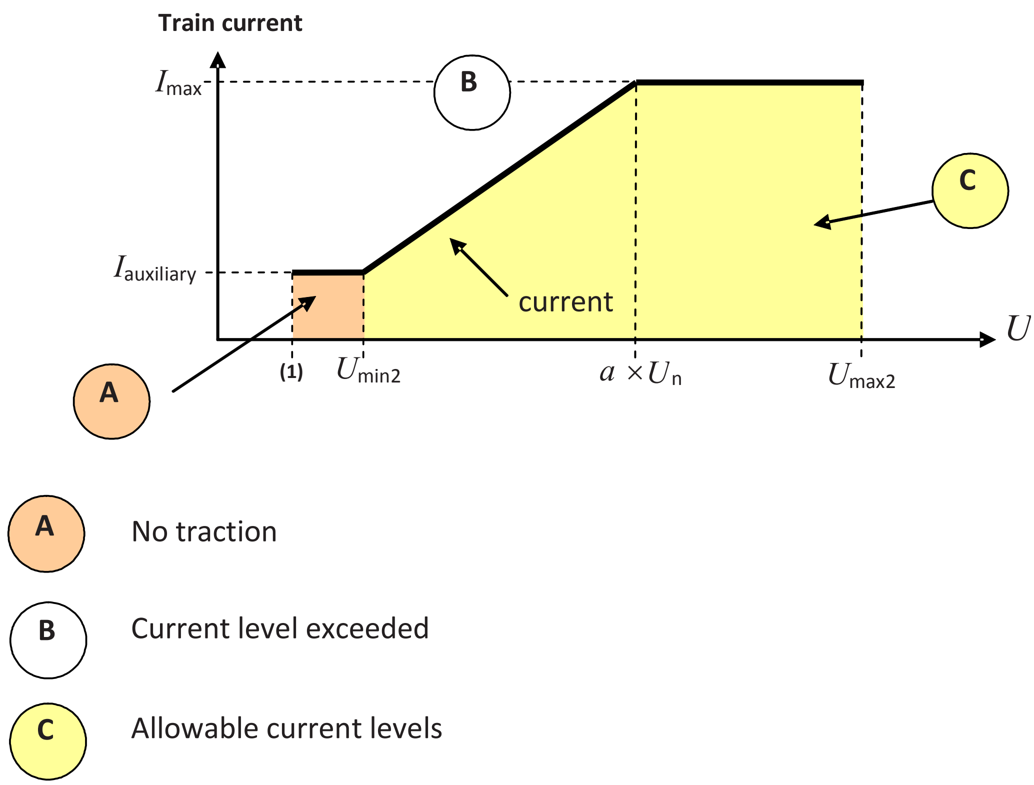
Typical data for the maximum allowable current per train for all European networks are given in informative Annex F.

NOTE There is no obligation to design the infrastructure to these maxima, e.g. allow 800 A in 25 kV (example).

7.2 Automatic regulation

In order to facilitate stable operation on weak power supply networks or in abnormal operating conditions, trains shall be equipped with an automatic device which adapts the level of the maximum power consumption depending on contact line voltage in steady state condition. Figure 1 gives the maximum allowable train current as a function of the contact line voltage.

Figure 1 does not apply in regenerative braking mode.



Key

U contact line voltage according to EN 50163

I_{max} is the maximum current consumed by the train at nominal voltage.

(1) With regard to the setting values of the under-voltage releases, see EN 50163:2004, 4.1, Note 2.

Figure 1 — Maximum train current against voltage

The value of the knee point factor a is given in Table 2.

NOTE The purpose of this diagram is not to design the nominal power of the train.

Table 2 — Value of factor *a*

Power supply system	Value of *a*
a.c. 25 000 V 50 Hz	0,9
a.c. 15 000 V 16,7 Hz	0,95
d.c. 3 000 V	0,9
d.c. 1 500 V	0,9
d.c. 750 V	0,8

7.3 Power or current limitation device

In order to allow a train to operate on both electrically weak and well supplied lines, on board current or power selection devices shall be installed which will limit the power demand of the train to the electrical capacity of the line. This is applicable on all lines except on category I lines.

The infrastructure manager shall declare the required limitation of each line in a register of infrastructure.

NOTE This setting may be carried out automatically.

7.4 Acceptance criteria

The train current limitation systems or devices are acceptable if the requirements of 7.1 to 7.3 are fulfilled.

8 Requirements for performance of power supply

8.1 General

A dimensioning study shall be performed in order to assess the ability of the power supply system to achieve the specified performance. Its aim, as set out in Annex B, is to define the characteristics of fixed installations.

These fixed installations should allow the most severe conditions, as specified in the design timetable, to be satisfied for:

– the densest operating period in the timetable, corresponding to peak traffic;

– the characteristics of the different types of train involved, taking account of the selected traction units.

The quality index $U_{\text{mean useful}}$ described in 8.2, is calculated by simulation and can be verified by ad hoc measurements on a critical train.

The voltage at the pantographs in EN 50163 and the $U_{\text{mean useful}}$ are relevant to this assessment.

8.2 Description

Mean useful voltage is calculated by computer simulation of a geographic zone (zone under study) which takes account of all trains scheduled to pass through the zone in an appropriate period of time corresponding to the peak traffic period in the timetable. This given period of time shall be sufficient to take account of the highest load on each electrical section in the geographic zone.

Account shall be taken of the electrical characteristics of the infrastructure and each different type of train in the simulation.

The fundamental voltage at the pantograph of each train in the geographic zone is analysed at each simulation time step. For a.c. systems, the r.m.s. value of the fundamental voltage is used. For d.c. systems, the mean voltage is used. The time step in the simulation shall be short enough to take into account all events in the timetable.

The voltage values from the simulation are used to study:

a) $U_{\text{mean useful}}$ (zone)

This is the mean value of all voltages analysed in the simulation and gives an indication of the quality of the power supply for the entire zone.

All trains in the geographic zone, over the peak traffic period considered, are included in this analysis whether they are in traction mode or not (stationary, traction, regeneration, coasting) at each simulation time step.

b) $U_{\text{mean useful}}$ (train)

This is the mean value of all voltages in the same simulation as the geographic zone study but only analysing the voltages for each train in the simulation at each time step where the train is taking traction load (ignoring steps when the trains are stationary, regenerating or coasting).

The mean value of these voltages gives a check on the performance of each train in the simulation and as a result, identifies the dimensioning train, i.e. the train whose ability to accelerate is most constrained by weak voltage.

8.3 Values for $U_{\text{mean useful}}$ at the pantograph

The minimum values for mean useful voltage at the pantograph under normal operating conditions shall be as given in Table 3.

Table 3 — Minimum $U_{\text{mean useful}}$ at pantograph

	Minimum mean useful voltage $U_{\text{mean useful}}$ at pantograph V	
Power supply system	**Category I, II, III HS TSI lines**	**Category IV, V, VI, VII CR TSI lines and Classical lines**
	Zone and train	**Zone and train**
a.c. 25 000 V 50 Hz	22 500	22 000
a.c. 15 000 V 16,7 Hz	14 200	13 500
d.c. 3 000 V	2 800	2 700
d.c. 1 500 V	1 300	1 300
d.c. 750 V	N.A.	675
Key N.A.: Not applicable		

8.4 Relation between $U_{\text{mean useful}}$ and U_{min1}

The power supply shall be electrically designed in such a way that the simulations of $U_{\text{mean useful}}$ under normal operating conditions never generate voltage values (r.m.s., mean values) at the pantograph of any train lower than the limit U_{min1} for traffic corresponding to the type of line concerned (see Clause 9).

8.5 Acceptance criteria

The electrification system design shall guarantee the ability of the power supply to achieve the specified performance in 8.3 and 8.4. Therefore, it shall be considered acceptable if the requirements of 8.3 and 8.4 are fulfilled.

9 Type of line and electrification system

The infrastructure manager shall declare the line parameters in terms of:

– line speed;

– minimum possible headway;

– maximum train power.

The power supply systems that are in use depending on the category of line are shown in Table 4.

Table 4 — Electrification systems as a function of the type of line

Range of speed, v	Categories of HS TSI lines		Categories of CR TSI lines and Classical lines	
km/h	I	II and III	IV, V, VI, VII	classical
300 ≤ *v*	a.c. 25 000 V 50 Hz; a.c. 15 000 V 16,7 Hz [a]	N.A.	N.A.	N.A.
250 ≤ *v* < 300	a.c. 25 000 V 50 Hz; a.c. 15 000 V 16,7 Hz [a]; d.c. 3 000 V [c]	N.A.	N.A.	N.A.
200 ≤ *v* < 250	N.A.	a.c. 25 000 V 50 Hz a.c. 15 000 V 16,7 Hz d.c. 3 000 V d.c. 1 500 V [b]	a.c. 25 000 V 50 Hz; a.c. 15 000 V 16,7 Hz; d.c. 3 000 V d.c. 1 500 V	a.c. 25 000 V 50 Hz; a.c. 15 000 V 16,7 Hz; d.c. 3 000 V; d.c. 1 500 V
v ≤ 200	N.A.	a.c. 25 000 V 50 Hz; a.c. 15 000 V 16,7 Hz d.c. 3 000 V; d.c. 1 500 V	a.c. 25 000 V 50 Hz a.c. 15 000 V 16,7 Hz d.c. 3 000 V d.c. 1 500 V	a.c. 25 000 V 50 Hz a.c. 15 000 V 16,7 Hz; d.c. 3 000 V; d.c. 1 500 V [d]; d.c. 750 V [e]

Key
N.A.: Not applicable

[a] See HS TSI Energy version 2008, Table 4.2.2, (1).

[b] For FR.

[c] For BE, ES, IT and PL see HS TSI Energy version 2008, Table 4.2.2, (2).

[d] For DK, d.c. 1 500 V has inverted polarity.

[e] For GB.

10 Harmonics and dynamic effects

10.1 General

The harmonic characteristics of the power supply and rolling stock in the railway system determine overvoltages in the power supply circuit (contact line and return circuit) and stability in this system. In order to achieve electrical system compatibility under steady state and dynamic conditions, harmonic overvoltages shall be limited below critical values. With protection devices installed, overvoltages cause an interruption of normal operation and are more critical from the operational point of view than from safety and security aspects. Annex C gives more information on these phenomena.

The following physical effects cause overvoltages.

a) Overvoltages caused by system instability:

Modern railway vehicles with inverter propulsion and auxiliary systems as well as static frequency converters are generally active devices which are capable of transferring energy from one frequency component in the spectrum to another. Their transfer behaviour is largely determined by the inverter controllers as well as by the passive elements.

Controllers have to be tuned or the impedance behaviour of the passive components (power supply network, vehicles) shall be optimised in such a way that for all operating conditions a stable behaviour results. In a non-stable system, physical values such as voltages and currents tend towards infinity and cause a shutdown in reality. In non linear systems, they can also continuously (steady state) oscillate on one or several frequencies.

Stability questions are always related to feedback loops in a system, especially through one or several controllers of one or several electrical subsystems. There is no explicit excitation source necessary to generate oscillation. This has to be distinguished from case b described below, where both an excitation source and a transmission/amplification path exist.

Normally the oscillations caused by instabilities are in the frequency range of up to about 1 000 Hz which is the bandwidth of the relevant controllers.

Low frequency oscillations below and around supply frequency are the result of the non-linear characteristics of modern converter systems. Higher frequency instabilities can be investigated within linear mode.

b) Overvoltages caused by harmonics:

Static inverters, both phase angle controlled and forced commutated ones, installed on rolling stock or in fixed power supply installations produce current and voltage harmonics which can be represented by current or voltage sources in a simplified manner. Every type of converter generates a typical current or voltage spectrum. The converter, combined with passive elements such as transformers and filters, shows a source behaviour and a typical internal impedance.

All power supply systems have resonances, due to the resonances of transmission lines and cables, and some also due to passive filter components. This leads to an amplification of harmonics injected by converters into the power supply system. An amplification or partial suppression occurs both at the location of the converter, due to the line impedance seen from the converter, and between the converter's location and other locations in the network. It consists in a transfer behaviour of the power supply itself.

An amplification of harmonics may lead to significant overvoltages, either at the vehicle's location or at a completely different location in the network.

The supply system has resonance peaks due to its distributed parameters - inductance and capacitance per unit of length. The resonance behaviour is changed by the presence of traction units. These resonance peaks can cause huge resonant currents and voltages. The ratio between maximum and minimum current recorded along the contact line at specific resonance frequencies could exceed 100. For four-quadrant converter vehicles the harmonic currents at the pantograph of a vehicle can be increased about 3 times due to resonance phenomena in the power supply network.

c) Other phenomena:

Further technical phenomena, which have to be considered for the electrical systems compatibility between power supply and rolling stock, are:

– multiple zero-crossings,

– voltage spikes and sags, transients,

– variations of the phase of supply voltage,

– low frequency oscillations,

– stray d.c. current flow at a.c./d.c. interfaces.

From a conducted interference point of view, the following effects can become relevant:

– wheel slip/slide;

– auxiliary load;

– dynamic events;

– harmonics from the auxiliary converter;

– bad current collection e.g. ice or frost;

– gaps in power supply (d.c. conductor rail).

10.2 Acceptance procedure for new elements

An acceptance procedure shall be carried out for the introduction of each new element, as defined in 3.9, by undertaking a compatibility study as defined in 10.3.

Compatibility between existing traction units and existing power supply, and new element(s) shall be checked against the phenomena described in 10.1.

Involved bodies or parties are:

– the owner(s) of existing infrastructure or infrastructure manager,

– the operator(s) of existing traffic,

– the purchaser/owner of the new traction unit(s) or infrastructure equipment,

– the manufacturer supplier of the new traction unit(s) or infrastructure equipment.

A general specification for rolling stock or power supply, which avoids overvoltages in any situations, might be very conservative and impossible to fulfil. Therefore, a process as described in 10.3 shall be applied to check compatibility (compatibility study).

10.3 Compatibility study

The compatibility study, or compatibility case, is a process to demonstrate the compatibility of the new element (new rolling stock or the new infrastructure component) with the existing traction units and power supply network. This process is shown in Figure 2 and explained in Table 5.

The amount of work to be done depends on the risk of the integration of the new element in view of the phenomena described in 10.2. If, based on earlier experiences, the integration is without any risk, this shall be documented in the plan, which is the only process step in this case.

The flow chart is applicable to rolling stock and also to fixed power supply components.

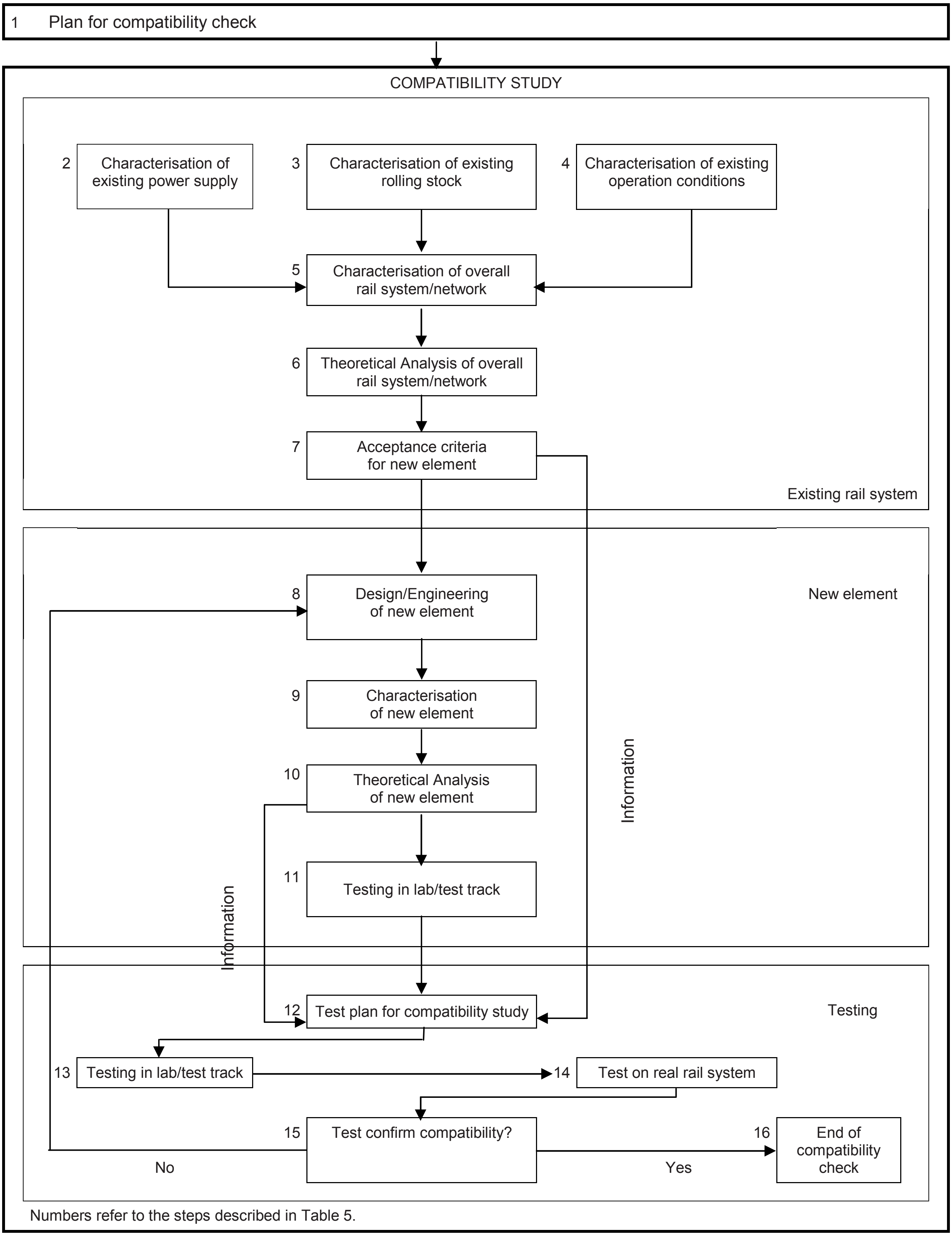


Figure 2 — Procedure for compatibility study of harmonics and dynamic effects

Table 5 — Description of steps (1 of 2)

No	Title	Description	Concerned party
1 [a]	**Plan for compatibility check**	The plan for a specific compatibility check when introducing a new element in an existing railway system defines the scope of the analysis, and the precise tasks and responsibilities. The plan shall be agreed between all parties involved. The organisation making the change shall be in charge of the compatibility study.	Organisation in charge of the compatibility check Plan to be agreed between the parties defined in 10.2
2	**Characterisation of existing power supply**	Characteristics of the existing power supply system, information relevant to the compatibility with vehicles (see Table D.1 and Table D.2). Normal feeding arrangements, emergency feeding arrangements.	Infrastructure manager
3	**Characterisation of existing rolling stock**	Characterisation of existing rolling stock operating on the network, information relevant to the compatibility with the power supply system information (see Table D.3 and Table D.4).	Operator / owner of rolling stock
4	**Characterisation of existing operation conditions**	Information about the operation of the existing system: number of trains in service, typical timetables, normal feeding arrangements, emergency feeding arrangements.	Infrastructure manager
5	**Characterisation of overall rail system / network**	This is the combination of the information from steps 2, 3 and 4. It may be necessary to define different scenarios.	Infrastructure manager
6	**Theoretical analysis of overall system / network**	Investigation of compatibility aspects for different scenarios. In a first step: confirm compatibility of the existing system. In a second step: test potential new element. check what characteristics they need to fulfil to maintain compatibility of the system.	Infrastructure manager
7	**Acceptance criteria for new element**	The result from the theoretical investigations in step 6 are the particular acceptance criteria for new element. The particular acceptance criteria must be understandable and measurable when designing and testing a new element. In future, these criteria shall be set up for all networks in identical form.	Infrastructure manager
8	**Design / Engineering of new element**	Design of new element, considering also the acceptance criteria defined in step 7.	Supplier of the new element
9	**Characterisation of new element**	The new element shall be modelled with respect to its compatibility with other vehicles and power supply elements. This model after validation in step 15 shall allow an amendment of the characterisation of the existing railway as required in steps 2 and 3.	Supplier of the new element
10	**Theoretical analysis of new element**	At an early stage of the design, it shall be checked in a theoretical analysis, e.g. using computer models, that the new element can meet the acceptance criteria.	Supplier of the new element
11	**Testing in lab / test track**	Once the first equipment (vehicle or power supply equipment) is built, it shall be tested in the laboratory or on a test track to verify that it meets the acceptance criteria as predicted by the theoretical analysis in step 10. This set of tests can be part of a type test of the new element.	Supplier of the new element

Table 5 – Description of steps (2 of 2)

No	Title	Description	Concerned party
12 [b]	**Test plan for compatibility study**	A plan shall be made to define the tests necessary to confirm as far as reasonably possible : 1) that the new element meets the acceptance criteria; 2) to make sure that the test reflects the reality. The tests shall demonstrate also the correctness of the simulations done in step 10 and shall be limited to critical cases.	Organisation in charge of the compatibility study To be defined in step 1
13 [b]	**Testing in lab / test track**	Tests will be performed in the laboratory or test track. These tests shall demonstrate that the acceptance criteria are met. A failure to meet the acceptance criteria means necessary modification of the new element by the supplier. This series of tests is part of a type test of the new element. The tests shall demonstrate also the correctness of the simulations done in step 10 and shall be limited to critical cases.	Organisation in charge of the compatibility study
14	**Tests on real rail system**	Tests on the real system shall give confidence that the acceptance criteria are sufficient to guarantee compatibility within the system after introduction of the new elements. If these tests show compatibility problems despite the compliance of the new equipment with the acceptance criteria, this means that the acceptance criteria were not sufficient. This set of tests is part of a type test of the new element. The tests shall demonstrate also the correctness of the simulations done in step 10 and shall be limited to critical cases.	Organisation in charge of the compatibility study
15	**Tests confirm compatibility**	If both sets of tests are successful, then compatibility of the new element with the existing system has been demonstrated. This shall be documented in a compatibility report.	Organisation in charge of the compatibility study
16 [c]	**End of compatibility check**	With the successful completion of the compatibility study, the new element becomes part of the existing railway system. The responsibility for its compatibility does no longer lie with the supplier of the new element.	Infrastructure manager

[a] The meaning of “new element” is given in 3.9.

[b] The test plan will define if both steps 13 and 14 or only one has to be performed.

[c] Seen from the compatibility point of view.

The result is a document describing the theoretical analysis and the measuring proof demonstrating that the vehicles and infrastructure are compatible in terms of conducted interference currents and stability.

10.4 Methodology and acceptance criteria

The overall acceptance criterion for overvoltages and stability is that no peak voltage higher than 30 000 V for 15 000 V, 16,7 Hz networks and 50 000 V for 25 000 V, 50 Hz networks will occur on the contact line in any point of the power supply network with voltage U defined in EN 50163 below or equal to U_{max2}. This value is the peak value of the distorted voltage waveform.

NOTE 1 For d.c. systems, no particular phenomena of system stability have been experienced. As a consequence, estimated values, are proposed as follows:

For 750 V use 1 300V

For 1 500 V use 2 600 V

For 3 000 V use 5 100 V

These values are derived from Table A.1 of EN 50124-2:2001.

These overall acceptance criteria are applicable in every case.

Since the overall acceptance criteria can only be applied to the complete power supply system including new element(s): it is beneficial to give design guidelines for the new element(s) which reduce the risk of failure in the compatibility study.

NOTE 2 The values above are only to define a criterion for identifying compatibility problems. They are not to define maximum allowable harmonic overvoltages for permanent operation. In case of long term operation, values in the above range lead to degradation of insulation and / or thermal damage in the infrastructure and vehicle components. Under short term operation, but with periodical appearance those voltages lead to a shortened lifetime of the components.

11 Coordination of protection

11.1 General

Protective systems on traction units and at substations shall be compatible.

11.2 Protection against short-circuits

The breaking capacity of the circuit breaker for a particular electrification system will determine whether or not faults can be cleared by the traction unit circuit breaker. Table 6 sets out the maximum contact line – rail short circuit levels. Table 7 sets out the action for internal faults within a traction unit.

Table 6 — Maximum contact line-rail short-circuit level

Power supply system	**Substation generally connected in parallel** Y/N	**Maximum fault current [c] that may occur** kA
a.c. 25 000 V 50 Hz	N	15 [a]
a.c. 15 000 V 16,7 Hz	Y	40
d.c. 3 000 V	Y	50 (prospective sustained [b])
d.c. 1 500 V	Y	100 (prospective sustained [b])
d.c. 750 V	Y	125 (prospective sustained [b])

[a] The value of 12 kA was previously commonly accepted.

[b] For definition see EN 50123-1:2003, 3.2.12.

[c] For the maximum currents, immediate tripping of the protection is assumed.

Key

Y = Yes

N = No

Table 7 — Action on circuit breakers at an internal fault within a traction unit

Power supply system	When any internal defect fault occurs within the traction units Sequence of tripping for:	
	Substation feeder circuit breaker	Traction unit circuit breaker
a.c. 25 000 V 50 Hz	Immediate tripping [a]	Immediate tripping
a.c. 15 000 V 16,7 Hz	Immediate tripping [a]	Primary side of the transformer: Tripping shall be staged [b] Secondary side of the transformer: Immediate tripping
d.c. 750 V, 1 500 V and 3 000 V	Immediate tripping [c]	Immediate tripping

[a] The tripping of the circuit breaker should be very rapid for high short-circuits currents. As far as possible, the traction unit circuit breaker should trip in order to try to avoid the substation circuit breaker tripping.

[b] If breaking capacity of the circuit breaker allows it, then tripping shall be immediate. Then, as far as possible, the traction unit circuit breaker should trip in order to try to avoid the substation circuit breaker tripping.

[c] When the current of the short-circuit is very high, the tripping of circuit breakers in the substations should be very rapid, and thereby prevents the traction unit circuit breaker clearing faults on it.

NOTE 1 New and modernised traction units should be equipped with high speed circuit breakers capable to break maximum short-circuit current in the shortest possible time.

NOTE 2 Immediate tripping means that for high level current, sub-station or train breaker should operate without introducing intentional delay. If first stage relay does not act, then second stage relay (back up protection relay) will act about 300 ms later. As information, with first stage relay, and by state of the art, duration of the highest short circuit current seen from the sub-station breaker is given hereafter:

- for a.c.15 000 V 16,7 Hz approximately 100 ms;
- for a.c.25 000 V 50 Hz approximately 80 ms;
- for d.c. 750 V, 1 500 V and 3 000 V approximately from 20 ms to 60 ms.

11.3 Auto-reclosing of one or more substation circuit breakers and effect of loss of line voltage and re-energisation on the traction unit

11.3.1 General

Auto-reclosing systems (where provided) for circuit breakers in the substation are liable to re-energise the line. In such a case, the substation circuit breakers shall not be re-closed before the tripping of the circuit breakers on the traction units present in the zone supplied by the substation.

The traction unit circuit breakers shall trip automatically within 3 s after loss of line voltage.

NOTE 1 For recommended values for undervoltage tripping, see EN 50163:2004, 4.1, Note 2.

On re-energisation, the traction unit circuit breaker shall not reclose within 3 s of the line being re-energised.

NOTE 2 The time delay on re-energisation allows for testing of the line for persistent short-circuit.

NOTE 3 In order to avoid a high level inrush current seen from the substation due to multiple simultaneous auto-reclosure of circuit breakers of identical type of traction units at standstill, it may be advisable to install on board a system that allows random time delays to successive auto reclosures within a range of hundreds of milliseconds.

11.3.2 Automatic reclosure after short circuits along the line

Automatic reclosure of the line circuit breakers is used to reenergize the line and to localize the affected line section.

Automatic reclosure is used directly or with line test via test resistor (or an electronic device). The following sequences shall apply:

a) Automatic reclosure with line test:

1) Tripping of the line circuit breaker;
2) waiting for all rolling stock in the line section for opening the on-board circuit breakers, to be done within 3 s after tripping of line circuit breakers;
3) 5 s after tripping of line circuit breakers, testing the line with line test via resistor (or an electronic device);
4) in case of a voltage recovery within the limits of EN 50163, the reclosure of the main circuit breakers shall be delayed for another 5 s (operational requirement for drivers) in order to ensure the line circuit breakers are switched on correctly.

b) Automatic reclosure directly:

1) Tripping of the line circuit breaker;
2) waiting for all rolling stock in the line section for opening the on-board circuit breakers, to be done within 3 s after tripping of line circuit breakers;
3) after at least 5 s reclosure of the line circuit breakers (in accordance with cycles defined in EN 50123 series for d.c.);
4) reclosure of the main circuit breakers of rolling stock when contact line is energized.

If rolling stock is operated in both protection philosophies, the reclosure of its main circuit breaker (step 4 above) should be as for 'reclosure with line test', i.e. with delay.

Where fitted, the line test system shall perform the test itself within 3 s.

11.4 D.C. electrification systems, transient current during closure

This provision is only applicable to d.c. traction units fitted with an input filter and without pre-charge device.

When the circuit breaker of a traction unit is going to be closed, with the input filter (if fitted), the transient current should not cause the protection devices in the substations to trip unnecessarily. The requisite information shall be obtained from the infrastructure managers concerned when vehicle-mounted filters are being designed.

The $\mathrm{d}i/\mathrm{d}t$ differential of the transient current on closure of the traction unit circuit breaker shall meet the following requirement, with a minimum contact line and substation inductance of 2 mH:

Regardless of the value of di/dt at closure of traction unit circuit breaker, di/dt shall reach a value lower than 20 A/ms within 20 ms after the time when di/dt reached 60 A/ms.

11.5 Acceptance criteria

Traction units shall be designed in accordance with the requirements as given in Tables 6 and 7 and in 11.3 and 11.4 (for d.c.).

The power supply system shall achieve the requirements given in 11.3.

12 Regenerative braking

12.1 General conditions on the use of regenerative braking

12.1.1 Traction unit conditions

Trains shall not continue to regenerate to the contact line if:

– there is a loss of supply voltage or a contact line-rail/earth short-circuit on the same section fed by the substation;
– the line voltage is higher than U_{max2} (see EN 50163:2004, 4.1);

– the contact line fails to absorb the energy.

If feedback energy absorption by other consumers is not available, rolling stock shall revert to other brake systems.

12.1.2 Power supply system conditions

For a.c. lines, the power supply system shall be designed to accept regenerative energy from the trains.

For d.c. lines, the power supply systems may be designed to accept regenerative energy from the trains. When the energy cannot be absorbed by other railway consumers, the infrastructure manager may then ask the power supply authority to accept the feedback of braking energy into the supply network or implement any other system able to absorb the energy.

Where it is not permissible to regenerate, it shall be recorded in the Infrastructure Register.

12.2 Use of regenerative braking

Table 8 shows where the regenerative braking mode is currently permitted by the infrastructure managers, on existing European networks.

Table 8 — Use of regenerative braking

Power supply system	Category of HS TSI lines		Category IV, V, VI, VII CR TSI lines and Classical lines																									
	I	II and III	Future system	AT	BE	CH	CZ	DE	DK	ES	FI	FR	GB	GR	HU	IE	IS	IT	LU	MT	NL	NO	PL	PT	SE	SI	SK
a.c. 25 000 V 50 Hz	Yes	Yes	Yes	/	Yes	/	No	/	Yes	Yes	Yes	Yes [a]	Yes [a]	Yes	?	/	/	/	Yes	/	Yes	/	/	Yes	Yes	/	No
a.c. 15 000 V 16,7 Hz	Yes	Yes	Yes	Yes [a]	/	Yes	/	Yes	/	/	/	/	/	/	/	/	/	/	/	/	/	Yes [a]	/	/	Yes	/	Yes
d.c. 3 000 V [b]	[b]	[b]	Yes	/	Yes	/	No	/	/	Yes	/	/	/	/	/	/	/	Yes	Yes	/	/	/	[b]	/	/	Yes	No
d.c. 1 500 V [b]	/	[b]	Yes	/	/	/	No	/	Yes	/	/	Yes	/	/	/	Yes	/	/	/	/	Yes	/	/	?	/	/	Yes
d.c. 750 V [b]	/	N.A.	Yes	/	/	/	/	/	/	/	/	/	Yes	/	/	/	/	/	/	/	/	/	/	/	/	/	/

Key
N.A.: Not applicable

[a] The value of the regenerated power in some places is limited due to old installations.

[b] In d.c. systems, by request of the train operator, the infrastructure manager can decide on the acceptance of regenerative braking.

12.3 Acceptance criteria

Traction units shall comply with the requirements given in 12.1.1.

13 Effects of d.c. operation on a.c. systems

On a.c. tracks that are in parallel with d.c. tracks and at system separation sections, there is a possibility of d.c. current flowing through the contact line of the a.c. system. This can lead to d.c. current flowing through the transformers connected to the a.c. contact lines, both of traction units and of the traction power supply system.

The d.c. current flowing in the a.c. installations mainly results from the operating currents and the rail to earth insulation of d.c. return circuit. In order to limit the d.c. currents in the a.c. system to an acceptable value, the requirements given in EN 50122-2:2010, Clauses 5 to 9 and EN 50122-3:2009, 8.3.2 and 8.4 shall be fulfilled.

NOTE 1 A d.c. component of 1 A in the transformer of a.c. traction unit has shown no adverse effect on vehicle performance.

NOTE 2 In ice conditions, arcs between contact wire and pantograph can result in a d.c. component in traction currents. Measurements have shown that short time d.c. currents can flow through the primary of the traction transformer of the locomotive or traction unit in the range of 0 A to 20 A. No adverse effect on vehicle performance has been noticed.

NOTE 3 d.c. components may trip train protection due to magnetising currents

The values mentioned above in Notes are informative and are not subject to assessment.

14 Tests

The tests given in Table 9 are applicable.

Table 9 — Tests

Title	Technical requirement clause	Traction units	Fixed installations	Test methodology subclause
Separation sections	5	X	X	15.1
Power factor of a train	6 (Table 1)	X		15.2
Train current limitation	7	X		15.3
Quality index of the power supply	8		X	15.4
Harmonics and dynamic effects	10	X	X	15.5
Coordination of protection	11	X	X	15.6
Regenerative braking	12	X	X	15.7

15 Test methodology

15.1 Separation sections

15.1.1 Tests for traction unit

Design review: Compliance with the sequences described in 5.1 and 5.2 shall be demonstrated by checking the control command diagrams of the traction unit at design phase.

Routine test (or site acceptance test): Tests on a line which is equipped on the infrastructure side to provide control signals to the traction unit shall be performed to demonstrate compliance with the sequences described in 5.1 and 5.2.

15.1.2 Tests for infrastructure

At the design phase, compliance with the functional requirements described in 5.1 and 5.2 shall be demonstrated by checking the electrical diagrams, at traction voltage level and control command level.

15.2 Power factor

Type test for traction unit: At the validation phase of the traction unit, on line measurements are performed at any point of curve tractive effort versus speed. The consumed power including auxiliaries, the traction or braking modes shall be recorded. Sufficient data shall be obtained in order to verify Table 1.

The values of the power factor shall be checked on board the train with a feeding system not limiting the performance of the train. The results of the on line measurement shall comply with 6.2 and 6.3.

15.3 Train current limitation

Type test for traction unit: The tests or checks shall be performed:

- at the design phase, by checking the functions of the control command of the traction unit and
- at the validation phase, by on line measurement. If necessary, outage of substations may be initiated to achieve low contact line voltage.

The results of the on line measurement shall comply with the requirements given in 7.2 and 7.3.

15.4 Quality index of the power supply

Table 10 — $U_{\text{mean useful}}$ (zone)

What	When	How	Acceptance condition
Simulation On a defined zone of the power supply system	After each simulation	Using the simulation results of the trains in the considered zone and calculation with definition in 8.2.	The value is greater than the values given in the relevant column “Zone and train” of Table 4.

Table 11 — $U_{\text{mean useful}}$ (train)

What	When	How	Acceptance condition
Simulation For a defined train in the simulation timetable – mostly the dimensioning train	As a result of simulations	Using the simulation results of the train calculation with definition in 8.2.	The value is greater than the values given in the relevant column "Zone and train" of Table 4.
Ad hoc measurement On a train motive power unit. This measurement is done within the context of traffic as listed in Clause 9. The measurement shall be done on a critical or dimensioning train from the simulated service. Several measurements should be made on the same train, in order to verify the convergence of the results and their validity. As far as possible, the reference timetable is the one used for the simulation. If not, a new simulation with the actual timetable shall be made. NOTE These measurements could, for example, be made on a day by day basis.	In response to problems or for validation purposes	— voltage recorder devices for the fundamental frequency or — digital data loggers with a frequency range greater/equal 2 kHz. Averaging over 1 s — measurement period: traction period of train — calculation with formula hereafter. From the recorded data, the following values or curves shall be calculated. a) The curve of voltage against time b) The mean value U_m expressed as: $U_{\text{m}} = \frac{1}{N}\sum_{\text{i}=1}^{N} U_{\text{i}}$ where N is the number of measured time steps in traction periods This formula is used only for traction period.	The value of U_m is greater than the values given in the relevant column “Zone and train” of Table 4 and verify that the recorded values (mean value or r.m.s.) of the voltage at the pantograph are not lower than U_{min1}.

Table 12 — Relationship between $U_{\text{mean useful}}$ and U_{min1}

What	When	How	Acceptance condition
Simulation	After each simulation	Using the simulation results of each train considered in the zone, the test shall be made only if $U_{\text{mean useful}}$ at the pantograph is higher than the values given in 8.3.	Check that the voltage at the pantograph of each train is never lower than U_{min1} as given in 8.4.

15.5 Harmonics and dynamic effects

For this assessment, 10.3 and 10.4 give all necessary information.

The tests shall demonstrate also the correctness of the simulations done in step 10 and shall be limited to critical cases

The instrumentation system used for recording the peak value of the distorted voltage waveform shall have an adequate frequency range. The r.m.s. value of the power frequency voltage shall also be recorded accordingly with the peak value of the distorted voltage waveform.

15.6 Coordination of protection

15.6.1 Protection against short-circuits and action on circuit breakers

The logical sequences of the triggering of the opening of the traction unit circuit breaker shall be verified:

– at the design phase by checking the functional electrical diagram of the traction unit and

– on response to problems or for validation when the traction unit is in service, by making a short-circuit test on board and checking that, in accordance with Table 7, the different circuit breakers trip in the correct sequence.

15.6.2 Auto reclosing system in substations

Test in any new substation: When the tests before putting into service are performed, the requirements given in 11.3 shall be verified.

If any, the line test system shall perform the test sequence within 3 s.

15.6.3 Loss of line voltage and effect on traction units

Type test: At the design phase of a traction unit, the requirements given in 11.3 shall be verified by checking the electrical diagrams.

Routine test: Tests on line shall demonstrate the conformity to the requirements given in 11.3.

15.6.4 D.C. traction units, transient current during closure

Type test for the traction unit: At the design phase, the requirements given in 11.4 shall be verified by checking the electrical design of the traction unit.

At on line test phase with a minimum contact line and substation inductance of 2 mH, the traction unit shall perform the following sequence of tests:

1) check that the on board filter is discharged;
2) close the main circuit breaker and record the waveform of the main circuit breaker current.

The characteristics of this inrush current shall be in accordance with the values specified in 11.4.

15.7 Regenerative braking

15.7.1 Traction unit

Type test: At the design phase, compliance to the requirements given in 12.1.1 of the control command of the traction unit shall be verified.

Routine test: Before putting into service, compliance to the requirements given in 12.1.1 shall be verified.

Therefore, during tests on line, power supply fixed installations shall be energised or de-energised from the substation.

Tests on line shall demonstrate that during braking mode of the traction unit the value of U_{max2} at the pantograph is not exceeded.

15.7.2 Substation

The substation installations e.g. control and protection devices, shall allow feedback energy to the supplying network or by any other means that improve the receptivity of the line. This shall be assessed by analysis of the connecting diagrams.

Where required, a combined system test shall be undertaken to verify the correct operation of substation protection with a regenerative traction unit on line.

Annex A
(informative)

Integration periods over which parameters can be averaged

A.1 General

Train operators or infrastructure managers use parameters for:

- dimensioning computations,
- protection measures,
- planning,
- etc.

These are effective only if they are averaged over precisely defined time spans. The subject of this clause is the definition of the periods over which these parameters should be referenced with the aim of achieving a simple comparison of these parameters for the different sections of a single railway, or of different railways. The reference time periods are given in Table A.1.

A.2 Reference time period over which values can be averaged or integrated

Table A.1 — Integration periods

Function	Duration	Comments
Protection	< 100 ms up to 300 ms for a.c. < 30 ms up to 100 ms for d.c.	Clearance of short-circuit faults depending on the intensity of the short-circuit current.
Back up protection	200 ms to 3 s	Clearance of fault on back-up protection.
Peak load current	1 s to 5 s	At supply points and for individual contact lines. Applicable to computer simulations, effect on feeding arrangements and assessment of back-up protection.
	1 min	r.m.s. average current for calculation of thermal loading on equipment.
Thermal overloading	30 s to 20 min preferred value: 10 min	r.m.s. average current for calculation of thermal loading on equipment.
Energy metering period	10 min, 15 min or 30 min	Average current or power for definition of the demand on the supply authority.
Long-term loading	1 h, 2 h	Conventional parameter to define long-term railway load.
Energy consumption	1 day, 1 week, 1 month, 1 year	Average energy consumption, used by electricity suppliers to study (for example) railway power requirements or when purchasing electricity.

Annex B
(informative)

Selection criteria determining the voltage at the pantograph

The aim of the dimensioning study is to define the characteristics of fixed installations.

These fixed installations should allow the most severe conditions, as specified in the timetable, to be satisfied through:

– the densest operating period in the timetable, corresponding to peak traffic;

– the characteristics of the different types of train involved, taking account of the selected traction units.

The design of fixed installations for electric traction can be obtained by simulating the critical timetable taking into account the power drawn by each train in the simulation at each time interval. Over and above aspects of sizing of equipment (transformers, contact lines, autotransformers e.g. 2 × 25 kV, and converters for d.c.) and compatibility with apparent performance tolerated at the high voltage connection points, the quality of the power supply constitutes an important qualifying parameter for the supply scheme studied.

The characteristic curve of tractive effort/speed for a traction unit changes as a function of the voltage at the pantograph. Determining the envelope of the characteristic tractive effort/speed curve under reduced voltage is achieved, in relation to the nominal characteristic curve, by extrapolation over the speed range, with the proportionality coefficient being slightly lower than the voltage ratio ($U_{\text{pantograph}}/ U_{\text{nominal}}$).

The voltage values obtained should allow the desired performance levels to be attained. For example, in order to study electrification with 25 kV, the selection of a voltage of at least 22,5 kV makes it possible, statistically not to drop below the minimum limit of 19 kV. Voltages below 19 kV are possible in periods of abnormal traffic with trains on closer headways in particular, or in the case of special situations that do not always appear in the simulations (coincidence of flights of traffic in directions 1 and 2).

The incidence of situations with impaired performance, both from the point of view of the power supply scheme and the train diagram, should be assessed with account taken of permitted losses in performance.

The selection of the correct mean useful voltage presents the following advantages:

1) it allows the traction units to function close to their nominal voltage (hence optimising efficiency and performance);
2) it ensures that the values of minimum voltage specified by the standards are respected;
3) it reflects the fact that the fixed installations for electric traction have appropriate additional capacity and that, as a result, increased traffic volumes can be considered;
4) it allows certain disturbed traffic situations to be dealt with.

The mean useful voltage at the pantograph can be defined mathematically as follows:

$$U_{\text{mean useful}} = \frac{\sum_{i=1}^{n} \frac{1}{T_{\text{i}}} \int_{0}^{T_i} U_{\text{pi}} \times \left| I_{\text{pi}} \right| \times dt}{\sum_{i=1}^{n} \frac{1}{T_{\text{i}}} \int_{0}^{T_i} \left| I_{\text{pi}} \right| \times dt} \qquad \text{(B.1)}$$

where

T_{i} is the integration or study period on train number i;

n is the number of trains considered in the simulation.

For a.c. electrification:

U_{pi} is the momentary r.m.s. voltage at fundamental frequency at the pantograph of train number i;

$|I_{\text{pi}}|$ is the modulus of momentary r.m.s. current at fundamental frequency passing through the pantograph of train number i.

For d.c. electrification:

U_{pi} is the momentary average d.c. voltage at the pantograph of train number i;

$|I_{\text{pi}}|$ is the modulus of momentary average d.c. current passing through the pantograph of train number i.

Formula (B.1) represents the relationship between mean power calculated for the train/trains during their traction sequences and the corresponding mean current.

A similar result is obtained with the following formula which is more suitable for some programs:

$$U_{\text{mean useful}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{M \times N \times \Delta t}\sum_{j=1}^{N}\sum_{k=1}^{M}\left(U_{j,k}(t) \times \Delta t\right)\right) \qquad \text{(B.2)}$$

where

n is the number of trains considered in the simulation;

$U_{\text{j,k}}(\text{t})$ is the voltage (a.c.: r.m.s. value of fundamental wave, d.c.: mean value);

M is the number of calculation steps in the integration period;

N is the number of summation steps over the simulation;

Δt is the time during which each calculation step M is simulated.

NOTE Δt should be short enough to take account of all events in the timetable.

Formulae (B.1) and (B.2) above are used to study:

a) a geographic zone (i.e. the part of the network one wishes to study) in a given period of time, with account taken of all the trains passing through the zone, whether they are in traction mode or not (stationary, traction, regenerating, coasting); the value of $U_{\text{mean useful}}$ therefore acts as an indicator of the quality of the power supply for the entire zone;

b) the mean useful voltage at the pantograph of each train taken in isolation; only the traction periods of the train are taken into account.

   In this case, in Formulae (B.1) and (B.2), n is equal to 1.

   This value is given to check the performance of each train in the simulation and as a result to identify the dimensioning train.

Annex C
(informative)

Investigation of harmonic characteristics and related overvoltages

C.1 General

A large disturbance of the supply voltage is caused if the total impedance / admittance of power supply and existing vehicles have a negative real part and zero imaginary part at the feeding point for one specific frequency, called resonance frequency. This condition is sufficient from mathematical point of view. The imaginary part of the impedance can be created by the global power supply alone or by the resulting inductance of the supply line and substation transformer and the corresponding input capacitance of a four-quadrant converter vehicle with primary input filter, respectively. Important for the correct prediction of whether or not a complete system is unstable is the knowledge of the correct total damping factor at resonance frequency. The damping factor is negative at an instability. In principle, simultaneous oscillations at several frequencies could be possible.

C.2 Overvoltages caused by system instability

C.2.1 Compatibility criteria

Considering the traction unit as a single electrical entity (boundaries are pantograph(s) – wheels), the most suitable way is to formulate restrictions on the input admittance of the vehicle, including the effect of all controllers inside the vehicle. This can be done either by defining allowed regions for magnitude and phase, or better for real and imaginary part, the real part directly defining the damping and/or excitation contribution. This reflects that there are no instability sources but that instability is always a question of transfer behaviour. Requirements for low frequencies, where non-linear effects play an important role, will lead to additional requirements. Knowledge about these effects is still limited.

Defining requirements for the vehicle requires, as described above, that the internal stability criteria for the whole power supply system have to be defined first, and have been broken down correctly to the one-vehicle interface.

To avoid resonance instability, for frequencies greater than a given value (to be defined by the Infrastructure Manager), the vehicle must be passive, i.e.: $\mathrm{Re}(Y(f)) > 0$, with a phase value of $Y(f)$ in the range between -90° and + 90°.

C.2.2 Measurement and validation

On a one-vehicle level, tests can be done by simulation, tests in the lab and on test tracks. Measurement of the vehicle input admittance versus frequency requires a power supply where a test voltage of a variable higher frequency can be superimposed on the fundamental supply voltage. In general this is practicable by means of different possible installations and has already been done before. The measurement of a non-linear behaviour could be more difficult since it normally depends on the operation point of a vehicle.

Compatibility tests between one vehicle and the power supply can mainly show whether stability is maintained or not. Instabilities can be detected either by protective shutdowns or observation of unwanted oscillations. It is difficult to measure the distance towards the stability margin. Damping of transients can give an approximate figure for this.

Validation of power supply models is difficult because of the large amount of the system data required and is therefore restricted to normal operation conditions.

C.3 Overvoltages caused by harmonics

C.3.1 Compatibility criteria

In principle the same is valid as for stability: since all power supply networks have both an input and a transfer behaviour, all network internal issues (such as amplification of harmonics between different locations in the network) have to be solved first.

For the compatibility check the manufacturer of rolling stock should get the following information from the infrastructure manager:

– Internal harmonic voltage source of the power supply. This includes the harmonics generated by the power supply itself (substations) as well as those generated by other vehicles. This can be done by definition of typically expected and maximum allowed voltage spectra at the pantograph of the vehicle;

– Admittance versus frequency of the power supply, seen at the vehicle's position. The admittance is provided for relevant conditions of the power supply network (including existing vehicles). A link to the voltage spectrum source information has to be made (e.g. a certain source spectrum with high harmonic content could be relevant only in combination with a weak network admittance).

Traditionally the vehicle should not exceed certain limits of harmonic current versus frequency (harmonic envelope). The harmonic envelope is normally defined with respect to signalling equipment, a similar approach is possible with limitation of overvoltages. The more restrictive harmonic envelope should be applied.

C.3.2 Measurement and validation questions

Measurement of harmonic currents and voltages is state of the art. Simultaneous measurements at different locations allow some conclusions to be made. Also the transfer behaviour of the power supply network and the plausibility of the requirements put on the new vehicle can be quantified.

C.4 Examples

C.4.1 General

This section shows examples of voltage (and current) waveforms from experienced phenomena causing overvoltages. The phenomena are classified according to the description in the text in this annex and in Clause 10.

C.4.2 Overvoltages caused by system instability

a) Electrical resonance instability

A voltage component is superimposed on the fundamental a.c. voltage. The typical frequency range of oscillation is three times fundamental frequency and above. It is caused by interaction between four-quadrant converter locomotives and system resonances (supply system with or without vehicle filters).

Example

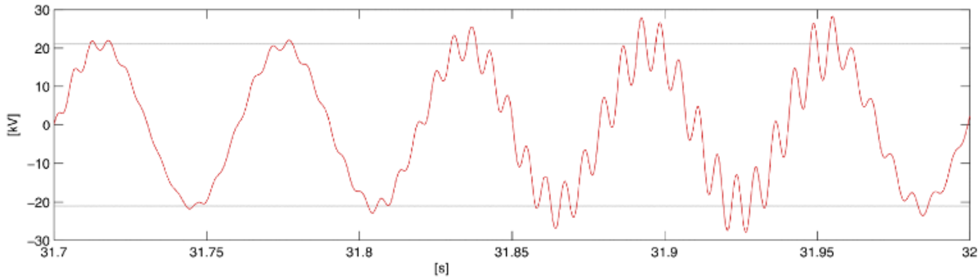


Figure C.1 – Line voltage

b) Low-frequency power oscillations

A.C. line voltage and current show an amplitude and phase modulation. The typical modulation frequency is 10 % to 30 % of the fundamental frequency. It is caused by interaction between four-quadrant converter vehicles and supply systems.

Example 1

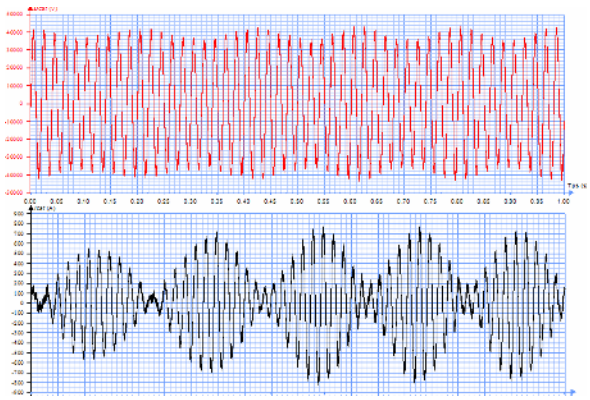


Figure C.2 – Line voltage (upper) and line current (lower) (instantaneous values)

Example 2

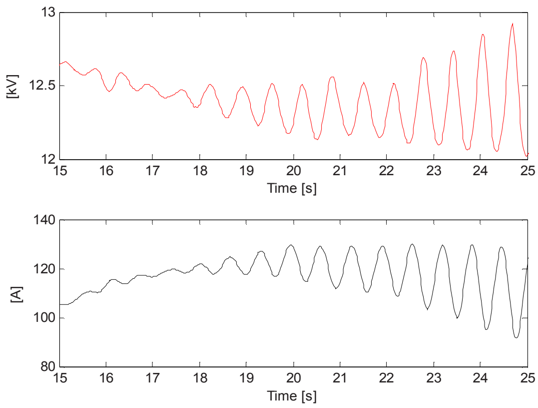


Figure C.3 – Line voltage (upper) and line current (lower) (RMS values)

C.4.3 Overvoltages caused by harmonics

Repetitive voltage distortions

a) Caused by commutations of thyristor rectifier vehicles at power supply resonances of higher frequency. Each peak in the voltage is followed by a transient waveform with relatively high damping.

Example 1

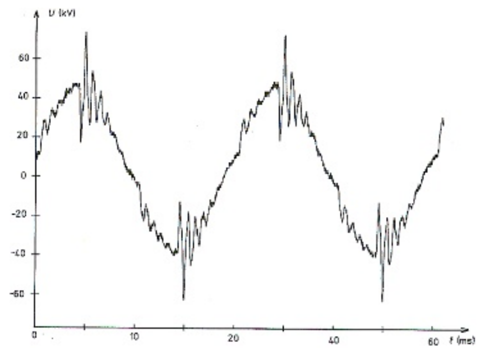


Figure C.4 – Line voltage

Example 2

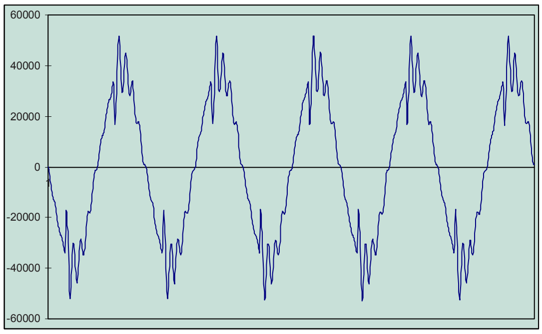


Figure C.5 – Line voltage

b) Caused by the switching frequency of pulse width modulated (PWM) converters. Multiples of the switching frequency fall together with a power supply resonance frequency and excite a strong oscillation on this frequency with relatively poor damping.

Example

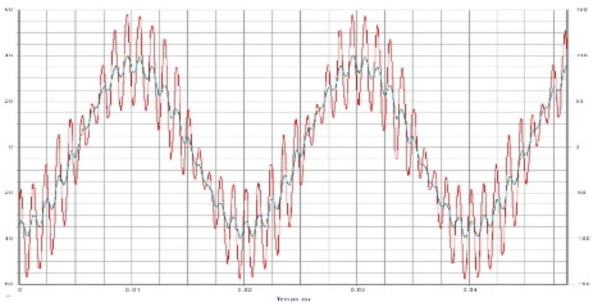


Figure C.6 – Line voltage

Annex D
(informative)

Data related to the compatibility study of harmonics and dynamic effects

D.1 Characterisation of the traction power supply fixed installations

To obtain complete and in-depth characterisation of the fixed installations for power supply is a huge effort. Furthermore, a general and simple characterisation for all types of fixed installations which is appropriate for the compatibility study, cannot be given, see 10.3.

However, as a guideline to typical situations and configurations of fixed installations, Table D.1 and Table D.2 describe the characteristics of a.c. and d.c. power supplies, mainly in terms of impedance, with their typical, maximum and minimum values from the point of view of the traction unit.

Particular values of systems which have different characteristics from those given in Table D.1 and Table D.2 should be given by the infrastructure manager.

The values given in Table D.1 and Table D.2 are valid for installations without significant length of cables at the supply voltage (50 m per sectors). In the other case, the infrastructure manager should indicate the location and characteristics of the cables.

Table D.1 — Characterisation of a.c. electrified lines

Type of line			Categories of HS TSI lines		Category IV, V, VI, VII of CR TSI lines and Classical lines		
			I	II, III	Power: High	Power: Medium	Power: Weak
Typical available power of the source	MW		20 to 60	15 to 45	15 to 45	15 to 30	10 to 20
Typical power of one train	MW		8 to 20	5 to 15	5 to 15	4 to 10	2 to 6
Length of the route supplied by one source in normal operating conditions. Distance from source to sectioning point [a]							
25 000 V - 50 Hz	km	min	15	20	15	15	20
		typ.	20	25	25	30	35
		max	30	30	30	35	40
2 × 25 000 V - 50 Hz	km	min	20	20	20	30	30
		typ	30	30	30	40	60
		max	45	50	50	50	70
15 000 V - 16,7 Hz	km	min	15	20	20	30	30
		typ	20	25	25	35	40
		max	30	35	35	40	60
2 × 15 000 V - 16,7 Hz	km	min	30	30	30	40	40
		typ	40	40	40	50	50
		max	60	60	60	70	70
Possible presence of compensation filters in the substation	Y/N		N	Y	Y	Y	Y
Tuned frequency of the filter: 25 000 V or 2 × 25 000 V - 50 Hz	Hz		N.A.	120 to 130 220 to 240	120 to 130	120 to 130 220 to 240	120 to 130 220 to 240
15 000 V or 2 × 15 000 V - 16,7 Hz	Hz			45	45	45	45
Quality factor 25 000 V or 2 × 25 000 V 15 000 V or 2 × 15 000 V			N.A.	100	100	100	100
Impedance of the source, fundamental frequency (high voltage grid, transformer) in: 25 000 V - 50 Hz	Ω	min	2	3	3	5	7
		typ	2,5	4	4	7	8
		max	3,5	7	7	10	10
15 000 V - 16,7 Hz	Ω	min	0,2	0,2	0,4	0,7	1,4
		typ	0,4	0,4	0,7	1,4	2,8
		max	1,4	1,4	1,4	2,8	4,5
Resonance frequency [c] of 16,7 Hz high voltage network	Hz	min	80	80	80	80	80
		typ	180	180	180	180	180
		max	300	300	300	300	300
Resonance frequency [c] of the electrified section of the route (without effect of filter(s) if any)							
25 000 V - 50 Hz	Hz	min	500	500	500	500	500
		typ	1 200	1 100	1 100	900	800
		max	1 400	1 200	1 200	1 200	1 000
2 × 25 000 V - 50 Hz	Hz	min	800	800	800	800	800
		typ	3 000	2 500	2 500	2 200	2 000
		max	3 500	3 500	3 500	3 500	3 000
15 000 V - 16,7 Hz	Hz	min	400	400	400	400	400
		typ	1 000	1 000	1 000	1 000	1 000
		max	2 800	2 800	2 500	1 400	1 300
2 × 15 000 V - 16,7 Hz	Hz	min typ max	Reserved				

Key
N.A.: Not applicable, Y: Yes, N: No

[a] When substations are connected in parallel, this length is half of the distance between two consecutive substations.

[b] Lω/R.

[c] Lowest resonance frequency.

Table D.2 — Characterisation of d.c. electrified lines

Type of line			Categories of HS TSI lines		Category IV, V, VI, VII of CR TSI lines and Classical lines		
			I [a]	II, III	Power		
					High	Medium	Weak
Typical available power of the source							
750 V			/	4	4	4	2
1 500 V	MW		/	8 to 15	8 to 15	4 to 8	2 to 4
3 000 V			10 to 20	6 to 12	6 to 12	3 to 6	3
Typical power of one train							
750 V			/	5	5	4	3
1 500 V	MW		/	7,5	7,5	5	4
3 000 V			8 to 12	6	6	3	3
Length of the route supplied by one source under normal operating conditions [b]							
		min		2	/	/	/
750 V	km	typ.		2,5	3	6	8
		max		3	/	/	/
		min		2	2	4	8
1 500 V	km	typ.		6	6	7	10
		max		8	8	10	10
		min	6	7,5	7,5	10	10
3 000 V	km	typ	8	10	10	12,5	12,5
		max	12	12,5	12,5	15	15
Possible presence of compensation filters in the substation (frequency)	Y/N			Y	Y	Y	Y
	Hz			600 1 200	600 1 200	600 1 200	600 1 200
			50 Hz (signalling purposes)				
750 V		GB		no	no	no	no
1 500 V	Hz	NL		no	no	no	no
1 500 V		FR		yes	yes	yes	yes
3 000 V		BE					
3 000 V		ES		600, 1 200	600, 1 200	600, 1 200	600, 1 200
3 000 V		IT	Aperiodic filters which attenuate f > 100 Hz	Aperiodic filters which attenuate f > 100 Hz	Aperiodic filters which attenuate f > 100 Hz	Aperiodic filters which attenuate f > 100 Hz	Aperiodic filters which attenuate f > 100 Hz
3 000 V	Hz	PL		600, 1 200 and aperiodic filters which attenuate either f > 100 Hz or f > 1 000 Hz	300, 600, 900, 1200 and aperiodic filters which attenuate either f > 100 Hz or f > 1 000 Hz	300, 600, 900, 1200 and aperiodic filters which attenuate either f > 100 Hz or f > 1 000 Hz	300, 600, 900, 1 200 and aperiodic filters which attenuate either f > 1 000 Hz or f > 100 Hz
3 000 V		SI				Aperiodic filters which attenuate f > 100 Hz	

[a] For those countries concerned by table footnote b of Table 5.

[b] When substations are connected in parallel, this length is half of the distance between two consecutive substations.

Key
Y: Yes
N: No

D.2 Characterisation of the trains

Table D.3 and Table D.4 describe respectively the characteristics of a.c. and d.c. traction units, mainly in terms of impedances, conducted interference harmonics and stability from the point of view of the traction power supply.

Particular values of vehicles which have different characteristics from those given in Tables D.3 and D.4 should be given by the operator(s) of existing traffic to the infrastructure manager.

Table D.3 — Characterisation of one a.c. train with respect to impedances, harmonics and stability (1 of 2)

Type of propulsion system	Tap changer	Diode rectifier	Thyristor rectifier	Four quadrant converter
Bandwidth of switching	3 Hz	3 Hz	2 × fundamental frequency	150 Hz, 6 000 Hz depending on switching device
Bandwidth of control	< 3 Hz	< 3 Hz	< 0,5 × fundamental frequency	1 000 Hz depending on switching device
Harmonic source				
Type of source	current source only due to transformer	current source due to switched large inductors	current source due to switched large inductors	voltage source on transformer secondary side due to switched d.c. link voltage
Type of harmonics	only odd natural harmonics	mainly odd natural harmonics	–mainly odd natural harmonics –some even natural harmonics during change of power or due to unsymmetrical firing angles	–depends on modulation method –usually mainly odd natural harmonics –few even natural harmonics –small interharmonics due to convolution effects in line inverter
Typical values for low frequency harmonics	very few due to transformer non linearity	large	large	only small, exist mainly if auxiliary frequency converter is of phase angle control type or due to regular sampling effects
Sub-harmonics	only few	only few	depends on adhesion and power control	depends on adhesion and power control
Medium frequency harmonics	low 50 Hz to 183 Hz (16,7 Hz) 150 Hz to 550 Hz (50 Hz)	medium 50 Hz to 2 000 Hz (16,7 Hz) 150 Hz to 2 000 Hz (50 Hz)	medium 50 Hz to 2 000 Hz (16,7 Hz) 150 Hz to 2 000 Hz (50 Hz)	medium 400 Hz to 10 000 Hz

Table D.3 — Characterisation of one a.c. train with respect to impedances, harmonics and stability (2 of 2)

Type of propulsion system	Tap changer	Diode rectifier	Thyristor rectifier	Four quadrant converter
Input admittance caused by control units				
Type of impedance for fundamental frequency	passive or active RL	passive RL	passive or active RL	passive or active R, RL or RC
Type of impedance for f > fundamental frequency	passive RL	passive RL	passive RL	active and/or passive RLC, depending on frequency and control
Non linearity	when motor fields saturated (at high traction effort)	significant due to switched impedance	significant due to switched impedance	mainly for low frequencies due to power and d.c. voltage control or inrush
Dependence on operation point	for all frequencies due to: – motor admittance depending on speed – transformation by tap changer	for all frequencies due to: – motor admittance depending on speed – transformation by tap changer – commutation angles depending on motor current	for all frequencies due to: – motor admittance depending on speed – firing and commutation angles depending on motor voltage	only for frequencies lower than the line frequency
Dependence on line impedance	none	yes, due to change of commutation angles with line impedance	yes, due to change of commutation angles with line impedance	none
Input impedance caused by filters				
Applied	not	partly	partly	partly
Resonance behaviour	Not applicable	100 Hz to 300 Hz	100 Hz to 300 Hz	250 Hz to 800 Hz 16,7 Hz 250 Hz to 800 Hz, 50 Hz
Key R is for resistance L is for inductance C is for capacitance				

Table D.4 — Characterisation of one d.c. train with respect to impedances, harmonics and stability

Type of propulsion system	DC motors with switched resistors	DC motors with chopper	AC motors with chopper + inverter	AC motors with direct inverter
Pulsing frequency (single component)	Not applicable	up to 300 Hz	up to 300 Hz	up to 5 000 Hz
Bandwidth of switching (complete converter)	3 Hz	400 Hz to 600 Hz or higher [a b]	400 Hz to 600 Hz or higher [b]	400 Hz to 600 Hz or higher [b]
Bandwidth of control	Not applicable [c]	50 Hz to 100 Hz	50 Hz to 100 Hz	50 Hz to 100 Hz
Input impedance caused by filters	Not applicable	> 0,3 Ω at 50 Hz	> 0,3 Ω at 50 Hz	> 0,3 Ω at 50 Hz
Resonance behaviour of filters	Not applicable	20 Hz to 40 Hz	20 Hz to 30 Hz	10 Hz to 30 Hz

[a] Frequency may be increased by a polyphase chopper. Frequency may be reduced at starting.

[b] Frequency can be increased up to few kHz when using IGBT.

[c] The starting resistors are excluded step by step if the currents is lower than the reference value.

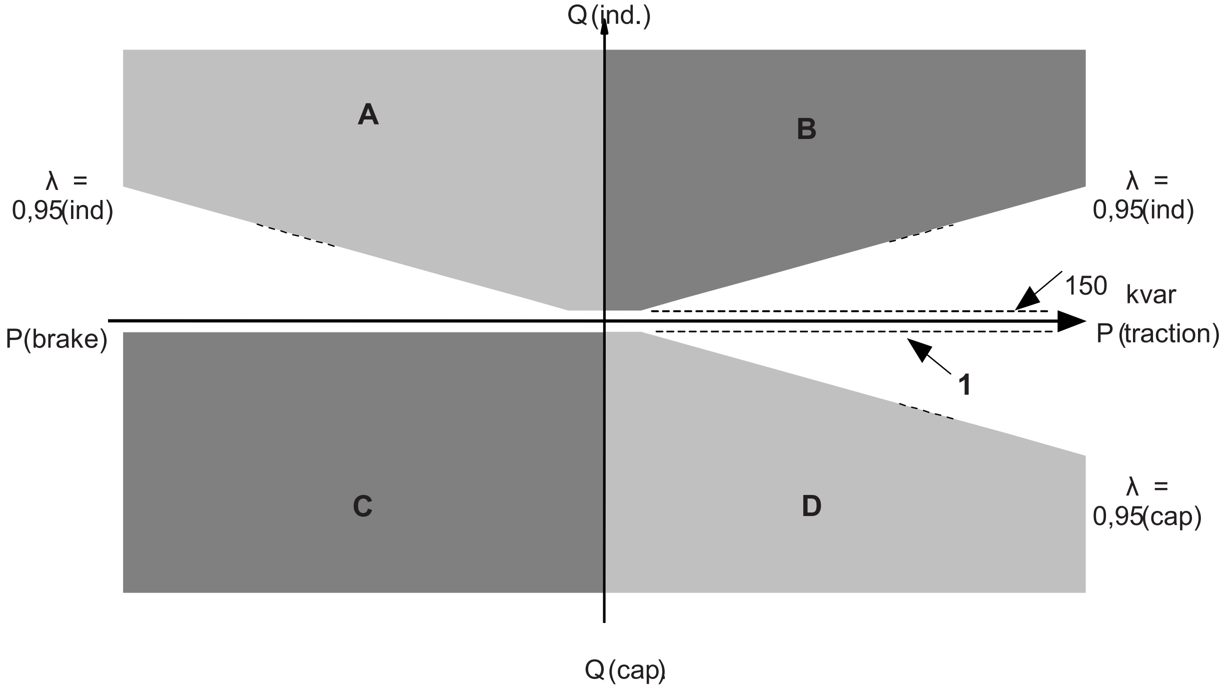
Annex E
(informative)

Total inductive and capacitive power factor

As an alternative to the requirements in Clause 6, Figure E.1 can be used for these requirements. Also in this Annex only the fundamental frequency is taken into consideration.

Following text is valid only for voltage below or equal to U_{max1} as defined in EN 50163.

The requirements represented in Figure E.1, are not mathematically strictly equal to the requirements in Clause 6, but for all practical cases these requirements will give the desired result. The requirements in this Annex are also fairly easy to verify by simple test runs. Meeting the requirements in this Annex will lead to fulfilment of Clause 6 for all practical cases.



Key

A is the undesired area for line voltages below normal feeding voltage;

B is the forbidden area for line voltages below $U_{\max1}$;

C is the forbidden area;

D is the undesired area for line voltages over normal feeding voltage;

P (traction) is the traction active power;

P (brake) is the braking active power;

Q (ind) is the inductive reactive power;

Q (cap) is the capacitive reactive power;

1 is the desired capacitive limit, see below for traction mode; for regenerative mode 150 kvar

NOTE All white areas around the P-axis are allowed without any considerations. Only the fundamental frequency is considered.

Figure E.1 — Allowed power factor versus drawn active and reactive power (P and Q) by the train

Note that for 50 Hz systems the desired limit of capacitive reactive power at low active power in tractive mode is 700 kvar and for 16,7 Hz systems the same desired limit is 150 kvar. The inductive limit at low active power is the same for both 50 Hz and 16,7 Hz systems: 150 kvar.

Figure E.1 is valid for a train in all normal situations, except for short durations (e.g. when going from tractive to braking mode and vice versa), when the power factor briefly can be out of limits, i.e. the figure is valid for all instantaneous r.m.s.-values under normal conditions, but transient states excluded. Also excluded are trains hotelling in yards or at depots, see below. Another exception is during icy conditions and low power consumption when a certain amount of inductive power (slightly more than 150 kvar) may be drawn temporarily in order to maintain continuous current.

At stand still or zero power, values on right side of the Q-axis should be used.

Furthermore the following is valid:

– it is not allowed to have a capacitive power factor at line voltages over U_{max1};

– for yards or depots the power factor of the fundamental wave should be equal to or greater than 0,8 (inductive) when the train is hotelling with traction power switched off and the drawn active power is greater than 200 kW.

NOTE 1 Even if the power factor is allowed to decrease freely (inductively) during regeneration (electrical braking), according to Clause 6, in order to keep the voltage within limits, it is desired that the power factor is not inductive (below 0,95) when the voltage goes below the normal feeding voltage.

NOTE 2 Capacitive power factors could lead to over voltages, exceedance of power transfer stability limits, or other dynamic effects, and should be treated in accordance with Clause 10. For low voltages, however, it could be desirable with a certain amount of capacitive power factor, provided that stability is maintained. Furthermore it is allowed but not desired to use capacitive power factor at line voltages over the normal feeding voltage. The reason that it is not desired is to secure the electrical brake capability of other vehicles in weak networks. Note that as soon as the traction inverters on the train are running it is possible to compensate for any eventual filters.

NOTE 3 The limit 150 kvar (inductive) corresponds to 200 kW at a power factor of 0,8.

Annex F
(informative)
Maximum allowable train current

The maximum allowable train current including auxiliaries for existing European networks, is given in Table F.1. The levels apply both in tractive and regenerative modes. Higher or lower values of train current shall be given in the register of infrastructure (see 7.3) for each line when required.

NOTE In order to prevent over dimensioning of the energy subsystem , the values given in Table F.1 are given for rolling stock and not for the design of the energy sub system for continuous load.

Table F.1 — Maximum allowable train current (1 of 2)

	Maximum allowable train current, A																									
Power supply system	**Category of HS TSI lines**			**Category IV, V, VI, VII CR TSI lines and Classical lines**																						
	I	II	III	Max.	AT CH DE	BE	CZ	DK	ES	FI	FR	GB	GR	HU	IE	IS	IT	LU	MT	NL	NO	PL	PT	SE	SI	SK
a.c. 25 000 V 50 Hz	1 500	600	500	800	/	500	800	500	?	500	500	300	/	?	/		/	?		500	/	/	?	500	/	300
a.c. 15 000 V [a] 16,7 Hz	1 500	900	900	900	900	/	/	/	/	/	/	/	/	/	/		/	/		/	900	/	/	900	/	/
d.c. 3 000 V [a]	4 000	4 000	4 000	4 000	/	2 500	3 000	/	2 500 (3 200 for HS TSI upgraded lines)	/	/	/	/	/	/		4 000	?		/	/	2 500 (3 200 for HS TSI upgraded lines; 2 500 for HS TSI connecting lines)	/	/	2 500	1 400 for single track line 2 000 for double track line

Table F.1 — Maximum allowable train current (2 of 2)

	Maximum allowable train current, A																									
Power supply system	**Category of HS TSI lines**			**Category IV, V, VI, VII CR TSI lines and Classical lines**																						
d.c. 1 500 V [a]	/	5 000	5 000	5 000	/	/	5 000	2 800	/	/	5 000	/	/	/	?		/	/		4 000	/		?	/	/	/
d.c. 750 V	/	/	6 800	/	/	/	/	/	/	/	/	6 800	/	/	/		/	/		/	/		/	/	/	/

[a] On special lines (e.g. freight in mountain area, suburban network), these values may be exceeded.

? no information received

Annex G
(normative)

Special national conditions

Special national condition: National characteristic or practice that cannot be changed even over a long period, e.g. climatic conditions, electrical earthing conditions.

NOTE If it affects harmonization, it forms part of the European Standard.

For the countries in which the relevant special national conditions apply these provisions are normative, for other countries they are informative.

Clauses Special national condition

7.2
Figure 1 **United Kingdom**

Where a line has not been upgraded to compliance with EN 50163:2004 + A1:2007, U_{min2} shall comply with United Kingdom special national condition to EN 50163:2004 + A1:2007, 4.1.

12.1.1 **United Kingdom**

For 750 V d.c. conductor rail systems, U_{max2} is replaced by 900 V.

12.1.1 **Netherlands**

Trains shall not continue to use their regenerative brake if:

— there is a loss of supply voltage or a contact line-rail/earth short-circuit on the same section fed by the substation,

— the contact line fails to absorb the energy,

— the line voltage is higher than U_{max2} (see EN 50163:2004, 4.1),

— the voltage is lower than 1 200 V d.c for 1 500V d.c. power supply and 17 500V for 25 000V a.c. power supply.

12.1.1 **Sweden**

U_{max2} is 17,5 kV instead of 18,0 kV.

Annex ZZ
(informative)

Coverage of Essential Requirements of EU Directives

This European Standard has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association and within its scope the standard covers all relevant essential requirements as given in Annex III of the EU Directive 2008/57/EC.

Compliance with this standard provides one means of conformity with the specified essential requirements of the Directives concerned.

WARNING: Other requirements and other EU Directives may be applicable to the products falling within the scope of this standard.

Bibliography

[1] EN 50119, *Railway applications — Fixed installations — Electric traction overhead contact lines*

[2] EN 50121 (all parts), *Railway applications — Electromagnetic compatibility*

[3] EN 50122-1, *Railway applications — Fixed installations — Electrical safety earthing and the return circuit — Part 1: Protective provisions against electric shock*

[4] EN 50124-2:2001, *Railway applications - Insulation coordination — Part 2: Overvoltages and related protection*

[5] EN 50238, *Railway applications — Compatibility between rolling stock and train detection systems*

[6] EN ISO 3166-1:1997 [1)], *Codes for the representation of names of countries and their subdivisions — Part 1: Country codes (ISO 3166-1:1997)*

[7] UIC 795, *Minimum installed power — Line categories*

[8] UIC 796, *Voltage at the Pantograph*

[9] UIC 797, *Coordination of electrical protection substations-traction units*

[10] UIC 798, *Integration periods over which parameters can be averaged*

1) Superseded by EN ISO 3166-1:2006, *Codes for the representation of names of countries and their subdivisions — Part 1: Country codes (ISO 3166-1:2006)*.

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